

Program Paper

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1 Goal (Current Thought)

1. Show that prolonged associativity operad $\mathbb{P}\mathcal{M}$ is well-defined as a G -operad.
2. Promote $\mathbb{P}\mathcal{M}$ to a well-defined permutativity G -operad in $\mathbf{Cat}_G$ (say $\mathcal{E}\mathbb{P}\mathcal{M}$), and use it to define the new pseudoalgebras.
3. Show that $\mathcal{F}_{G,\mathbb{R}}$ pseudofunctors (if possible, G -functors, meaning preserving morphism sets' G equivariance) is well-defined, together with isomorphism to new pseudoalgebras as above.
4. (Not sure if it's true), but if we can show there is an equivalence of G -categories between the original $\mathcal{P}_G$ and $\mathcal{E}\mathbb{P}\mathcal{M}$, then maybe we can also show the equivalence from the original categories.

The first three goals based on intuition and inspection seem to be true.